



Tracing global narratives: Uzbekistan's take on the world

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INTRODUCTION

Since news portals have become daily used by all individuals in the world, both official and traditional news media including other platforms such as social media, blogs, and discussion forums started to become crucial as a main venue of news. Not only can people access the news via different translation software, but they can also easily access different sources of information in their language. In an increasingly globalized world, foreign news is becoming more relevant than ever, thanks largely to real-time, online coverage of events and issues in remote settings.¹ To explore the phenomenon of international news, this paper analyzes the narratives of global news from the popular news media Kun. Kun.uz is a private and leading digital news platform in Uzbekistan, dedicated to providing timely and accurate news on politics, economy, society, culture, and international affairs. According to SimilarWeb, 7.4 million people visited the site in January 2023.² In this paper, the highlights of the period from 2018 to 2022 of foreign news coverage by the “Kun.uz” being analyzed and the findings are to answer the puzzles which relate to the pattern of foreign affairs in the field of internal alliances in the eyes of Uzbekistan population.

This analysis raised three key questions while building network structures to reveal insights into international connections, all with the same aim which is to explore the relation and interconnectivity between different countries. Firstly, we test the Uzbek public’s exposure to global geopolitics and interest in intergovernmental organizations. Secondly, we explore the patterns among the network of countries that are being co-mentioned to reveal the global narratives such as neocolonialism, Soviet legacy, and Westernization. Finally, we arrange the patterns of realignment or fragmentation to find the impact of the highest nodes/ countries and their dependency. The last tryout aims to conclude the balance of global news flow and to test ‘what if such nature of news has shifted to a direction we have never thought of’. As it is necessary to mention, all these outcomes from our analysis are to be considered as limited as our knowledge about the nature of political understanding is solely based on our interpretation. Drawing conclusions relating to different figures in this paper can be interpreted differently and the purpose of this paper is to explore the nature of the network among countries.

Motivation

We are interested in implementing computational social science knowledge in mass communication and journalism, specifically in international news coverage and online news organizations. Our goal is to analyze foreign news coverage in Uzbekistan and how these variations mirror larger socio-political, cultural, and economic factors. We also wonder to find out which stories are prioritized and how these findings might affect public perception. Our main input is to find insights that help create more inclusive journalism in developing countries.

¹ Philip Seib, *Going Live: Getting the News Right in a Real-time, Online World* (Lanham, MD: Rowman & Littlefield, 2002); Bruce Garrison, "Diffusion of Online Information Technologies in Newspaper Newsrooms," *Journalism 2* (August 2001): 221-39

² <https://kun.uz/en/news/2023/02/23/kunuz-is-the-most-popular-online-publication-in-uzbekistan-yuksalish>

DATA

Dataset Overview

For this study, a publicly available dataset from Kaggle³ was implemented. The dataset contains 172,349 news articles from Kun.uz, a leading news platform in Uzbekistan. The timeline spans from December 2018 to September 2022. These articles were obtained through web scraping by the AIRI Team (Digital Technologies and Artificial Intelligence Research Institute)⁴, a scientific organization under the Ministry of Digital Technologies of Uzbekistan focused on AI research.

The dataset has the following variables:

- News Title (title): The headline of the news article.
- News Content (content): The main body of the news article.
- News Type (target): The category of the news. Categories: Uzbekistan, World, Society, Sports, Business, Science and Technology, Economy.

About Kun.uz

Kun.uz is one of the most active Uzbek-language online news platforms, established in January 2012⁵. It is a private digital news company led by Umid Shermammedov with over 50 staff members⁶. It is officially registered under the State Press and Information Agency in Uzbekistan and delivers news in Uzbek, Russian, and English. The most active audience is on social media platforms: Instagram (4.7M followers), YouTube (1.7M followers), Telegram (1.1M followers), Facebook (573k followers), X/Twitter (163k followers)⁷. However, most of the activity is still centered on their website. According to Google Analytics, the website attracts around 351,000 daily readers and exceeds 10 million monthly visitors, with over 23 million monthly page views⁸. Kun.uz is also ranked as the 1st most popular news website in Uzbekistan by FeedSpot⁹.

METHODS

Article Pre-Processing

Only articles classified under the "World" category were chosen for analysis. Then, considering the unique Uzbek spellings of country names and organization titles, we created a comprehensive list of countries and intergovernmental organizations (IGOs). We identified 195 countries based on Wikipedia¹⁰ and chose following IGOs: United Nations, European Union, NATO, Shanghai Cooperation Organization,

³ <https://www.kaggle.com/datasets/airiteam/kun-uz-yangiliklar>

⁴ <https://airi.uz/en/>

⁵ <https://uz.wikipedia.org/wiki/Kun.uz>

⁶ <https://kun.uz/en/our-team>

⁷ <https://kun.uz/en/page/reklama>

⁸ <https://kun.uz/en/page/reklama>

⁹ https://news.feedspot.com/uzbekistan_news_websites/

¹⁰ https://en.wikipedia.org/wiki/List_of_countries_and_dependencies_by_area

G7, G20, BRICS, African Union, Arab League. To find references to countries and IGOs, we examined each article's content using text-matching and keyword detection. Regardless of their placement within the article, these mentions were identified to guarantee that all information was recorded. Lastly, for each article, we added two new columns to update the dataset: countries with a list of all the countries mentioned and alliances with a list of the intergovernmental organizations referenced.

Network of Co-Mentions

We constructed an undirected network to analyze relationships between countries based on their co-mentions in news articles. We generated edges by identifying every pair of co-mentioned countries that could exist in each article. For instance, if an article mentioned three countries—A, B, and C—the resulting edges were (A, B), (A, C), and (B, C). The network was designed in naïve way, meaning the order of countries in a pair and direction of edge was not considered. Countries are represented as nodes in the network, while co-mention relationships are shown by edges. The strength of the relationship between any two countries is the frequency of these co-mentions.

Visualization on a World Map

We used geographic data from Natural Earth¹¹, which offers comprehensive vector data for mapping at a worldwide scale, to display the network of country co-mentions on a map. With the use of edges that depict the relationships between countries in the media coverage, we were able to overlay the network of co-mentions onto a global map. By doing this, we added context for comprehending regional clusters and patterns by integrating the geographic proximity of nations into the display. This method makes it simpler to spot spatially driven trends in international news coverage by highlighting the links between nations.

Louvain Community Detection Algorithm

We used the Louvain algorithm to find groups of closely related nations based on their co-mention patterns in order to improve the network analysis. A popular approach for detecting communities, the Louvain method maximizes modularity by efficiently classifying nodes into communities with more connections within the group than those with nodes outside the group.

We distinguished each identified community on the map by giving it a distinct color by including the Louvain clustering findings into the visualization. This made it possible for us to see both the specific relationships between nations as well as the more general structural trends in the way that countries are arranged within the network. The map's visualization of these clusters' physical proximity offers more information on how regional or thematic linkages affect media coverage. This method draws

¹¹ <https://www.naturalearthdata.com/downloads/110m-cultural-vectors/>

attention to how countries are interconnected in global narratives and identifies clusters that could stand for mutual interests, regional alliances, or geopolitical narratives.

Node Attacks

We used a node attack methodology to test hypotheses about dependency theory and the presence of core centers in international news coverage. This involved removing any article that mentions the three main geopolitical powers — China, Russia, and the United States — from the dataset and tracking how the network structure changed as a result. First, we eliminated all articles that mentioned any of them for the multiple-node attack. In order to evaluate each country's distinct influence on the network, we also carried out singular attacks by eliminating articles that mentioned that specific country. We recalculated the edges between co-mentioned nations from the remaining articles after filtering. This gave us a better understanding of key nodes in the global media ecosystem and allowed us to investigate how the absence of these key actors affected the network as a whole.

RESULTS

To analyze, we decided to create 3 studies.

Study 1: Overview

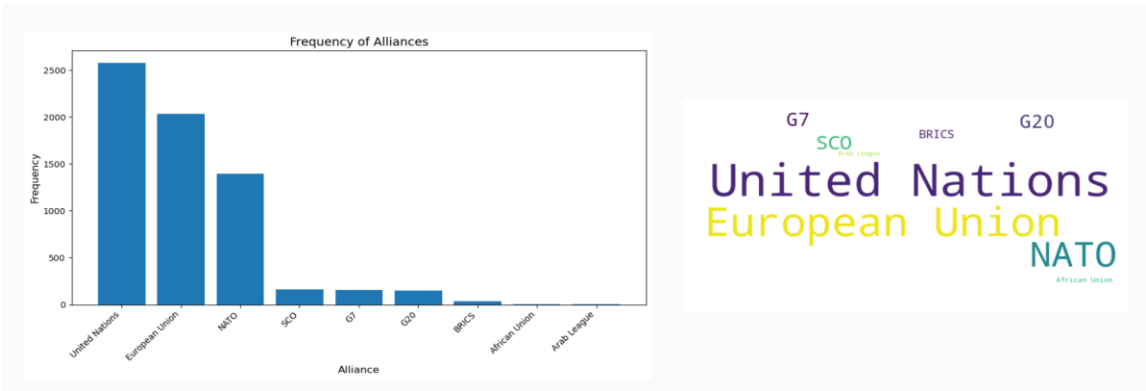


Figure 1. Visualization of Alliance Frequency and Prominence in International Networks

In Fig (1), the frequency and prominence of international alliances have been highlighted. The Bar Chart shows the frequency of mentions or associations for various alliances. The most frequently mentioned alliance is the United Nations, followed by the European Union and NATO, while groups like the Arab League and African Union are the least prominent according to our ranking. The word cloud highlights the significance of alliances, with larger text representing their frequency, as seen in the prominent display of the United Nations, European Union, and NATO, emphasizing their crucial role in international relations.

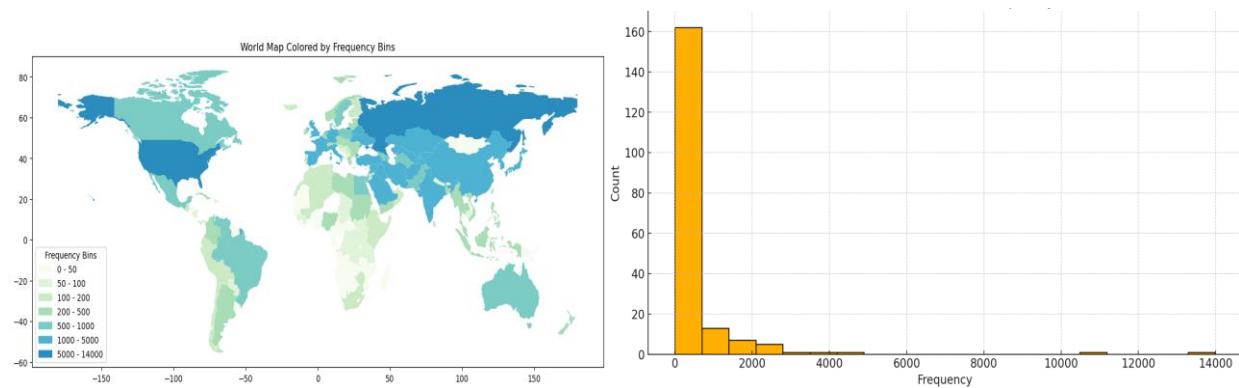


Figure 2. Geospatial Distribution and Frequency Histogram of Global News Coverage

According to the World System Theory, built upon the dependency theory¹², conceptualized by Wallerstein, shows a three-tier framework in which countries are integrated into a global structure. Based on historical, economic, and political factors, countries around the globe are classified into three interactive zones known as core, semi-periphery, and periphery.¹³ In Fig (2), regions like the United States, parts of Europe, and China appear to have higher values representing the core, meanwhile, most of Africa, South America, and part of Asia display lower intensity. This gradient provides a depiction of geographical variations, and we can thus further determine the core-periphery structure in the international news network of Uzbekistan.

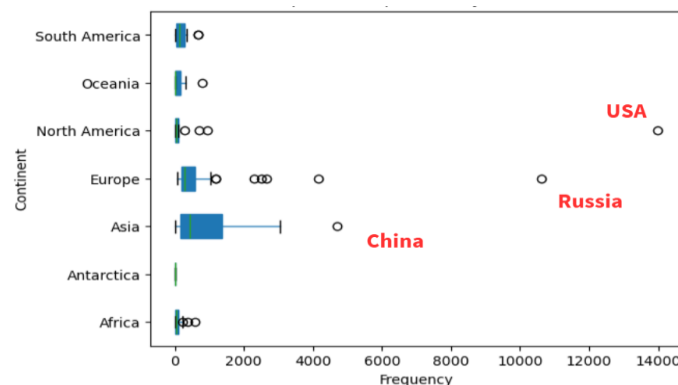


Figure 3. Boxplot of country mentions by continent with outliers

Relying on the discovery of the visualization of the heatmap is conceptual. However, plotting the data distribution to define the core can be more useful as we can separate the outliers' countries among different contents. As shown in Fig (3), the three main outliers are the People's Republic of China in Asia

¹² Jorge Larraín, *Theories of Development: Capitalism, Colonialism and Dependency* (Cambridge, UK: Polity Press, 1989); Alvin Y. So, *Social Change and Development: Modernization, Dependency, and World-system Theories* (Newbury Park, CA: Sage, 1990); Iain Wallace, *The Global Economic System* (London: Unwin Hyman, 1990).

¹³ Himelboim, I., Chang, T.-K., & McCreery, S. (2010). International Network of Foreign News Coverage: Old Global Hierarchies in a New Online World. *Journalism & Mass Communication Quarterly*, 87(2), 297-314. <https://doi.org/10.1177/107769901008700205>

(4694 mentions, 11.4% of articles), the Russian Federation in Europe (10610 mentions, 25.7% of articles), and the United States of America in North America (13982 mentions, 33.9% of articles).

Study 2: Network

Code in-degree distribution.

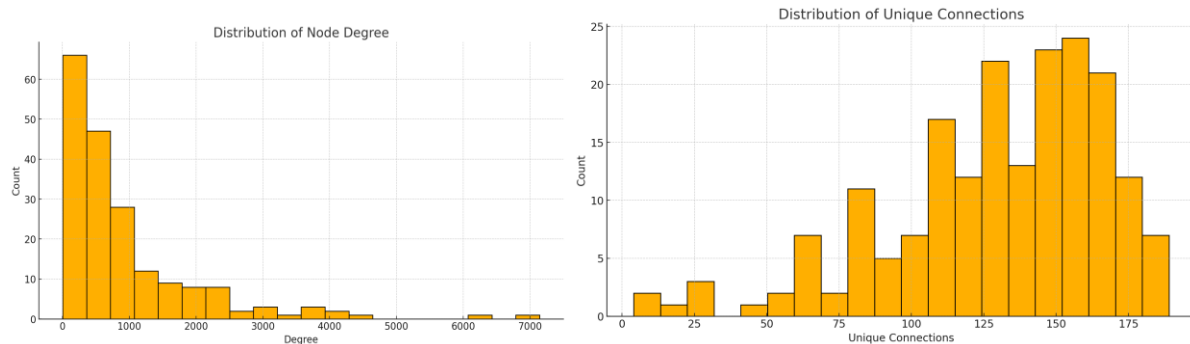


Figure 4. Distribution of Node Degrees Unique Connections in the Network

The two graphs provide critical insights into the structural characteristics of the network. The first graph, illustrating the distribution of node degrees, highlights a skewed pattern where most nodes have a low degree, while a small number serve as hubs with exceptionally high connectivity. This is indicative of a scale-free network, a structure commonly found in real-world systems. In contrast, the second graph, showcasing the distribution of unique connections, reveals a more balanced distribution, with most nodes maintaining a moderate number of distinct connections, peaking between 125 and 175. This suggests that while the network is hierarchical with influential hubs, there is also a degree of diversity in how connections are distributed, potentially facilitating a broader exchange of information across nodes. These graphs underscore the dual dynamics of centralization and diversity within the network, offering a nuanced understanding of its functionality and flow of interactions.

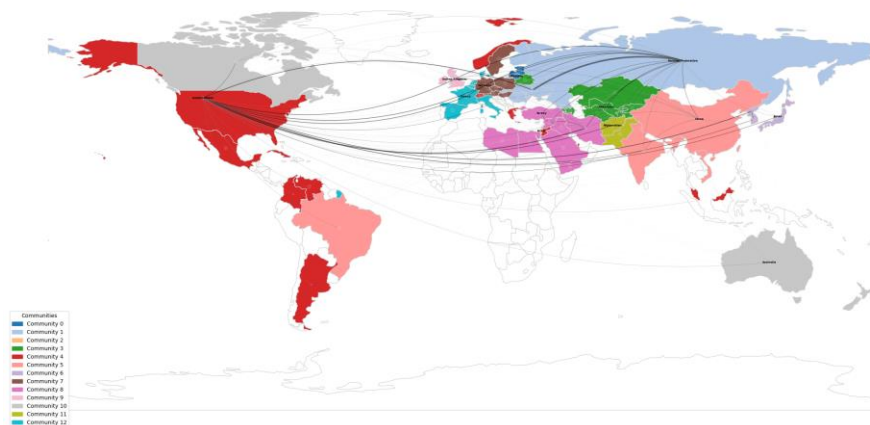


Figure 5. Louvain community detection (12 communities)

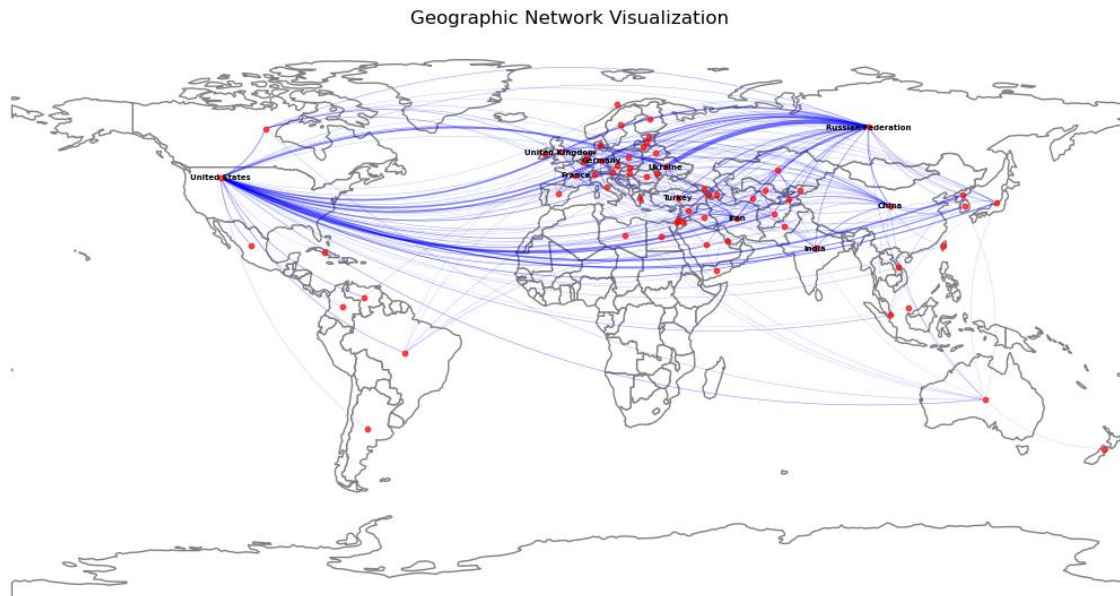


Figure 6. Triadic Network Structure

Based on the Louvain community detection, the world map of the explored international news flow can be reflected in the asymmetric patterns suggested by the World System Theory. Figure (5) illustrates the division of twelve distinct communities and the interaction concentration among core countries while peripheral countries remain more loosely connected, reinforcing the triadic structure proposed by Wallerstein. Core countries, such as the United States, those close distance with Russia, and China dominate the international communication network by fostering dense interconnections within their group while maintaining limited interactions with peripheral regions. Figure (6) portrays the nature of the three core countries in a clearer sense.

This structure mirrors the dynamics of global capitalism, where news agencies located in core countries are being prioritized for the coverage of events. Peripheral countries, on the other hand, tend to connect under the same region although our designed communities are not provided with the geographical knowledge, meaning that these given 12 communities are not programmed to understand which countries are from the region of the world. Due to the nature of these countries mentioned in the news, the regions are divided by themselves and portrayed in a regional division. West Europe tends to have more connections with regional collaboration or neighboring states for example, these structural patterns can be prevalent in central Europe, central Asia and the Middle East. The map highlights the need for further research to explore how these dynamics evolve over time and whether our Kun media can genuinely balance the asymmetrical flows of information in the global network.

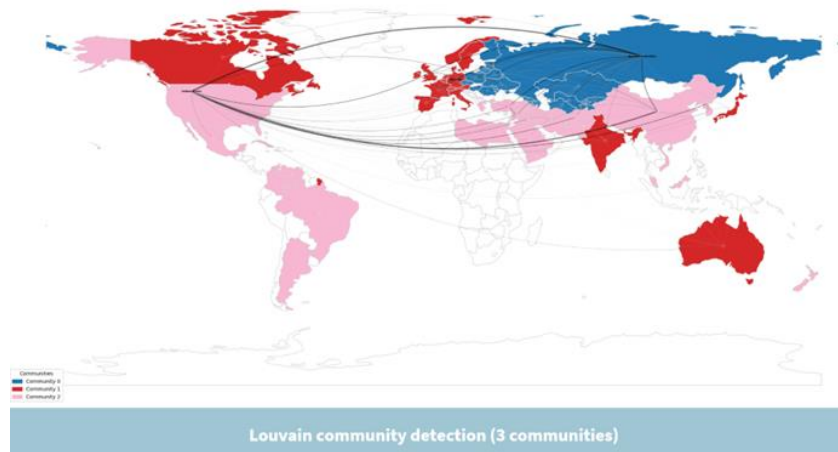


Figure 7. Three communities of Louvain community detection

In the context of neo-colonialism and the post-Soviet era, one might expect the dynamics of connections between former empires to be entirely dismantled and a completely new structure would appear. However, after analyzing the communities and reducing them to three clusters, our evidence suggests that remnants of the pre-existing geographical powers persist, portraying the lingering influence of historical empires. To prove our claim further, we divide the international network into three distinct communities, and the hierarchical and asymmetrical nature of global communication as outlined by the World System Theory can be found. The red community, predominantly representing core nations such as Australia, and much of Western Europe, India, and Canada, demonstrates dense interconnections, reflecting their dominance in shaping international discourse as they are the former colonial power from pre-world wars. The blue community, encompassing Russia and parts of Eastern Europe and Central Asia, highlights a legacy of the Soviet Union with significant regional interactions. Meanwhile, the pink community, largely composed of peripheral and semi-peripheral countries in the United States, Latin America, Africa, and parts of Asia, illustrates the connections of the nature of economic and political affiliations, often focused on interactions with core countries including China. This pattern aligns with Wallerstein's triadic structure, where core countries maintain dominance, semi-peripheral nations act as intermediaries, and peripheral nations remain marginal. The concentration of connections among core countries perpetuates information asymmetries, with major foreign news in core regions driving global narratives. Peripheral countries, despite the opportunities provided by the Internet, seem to miss chances to strengthen regional ties, likely due to economic dependencies and elitist perceptions of newsworthiness. This visualization highlights the enduring inequities in global news flows in Kun newsworthiness.

Study 3: Core periphery

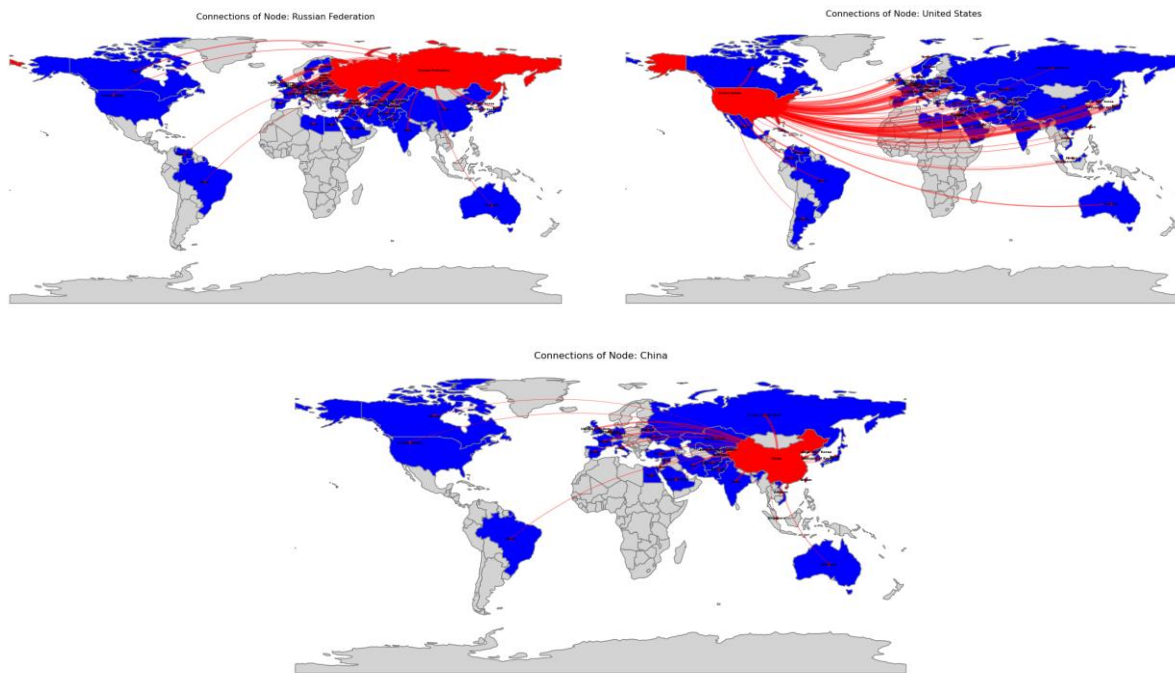


Figure 8. Connections of the Russian Federation, United States, and China

The global connections of the Russian Federation, the United States, and China, as depicted in Figure (3), highlight their central roles or are considered as “core” in geographic and network dominance, showcasing their pivotal influence in shaping international relations. According to Figure (8), The Russian Federation and The United States display strong regional ties, particularly with Eastern Europe, Central Asia, and parts of the Middle East, reflecting its historical and geopolitical influence in these areas. Its connections extend to a smaller number of global partners compared to China. The United States, on the other hand, demonstrates a much broader and denser network, with strong ties across Europe, the Americas, and parts of Asia-Pacific, indicating its central role as a global hegemon in political, economic, and communication networks. Meanwhile, China shows significant regional connections, especially in East Asia and Southeast Asia, with selective ties extending to Europe, Africa, and South America. China and its connection throughout central Europe are neglected, reflecting its influence as an emerging global power and its limitations. These figures highlight the distinct regional and global strategies employed by these nations, with the United States exhibiting the widest global reach, Russia focusing on regional dominance, and China balancing regional influence with strategic global connections. Without the influence of these three core-periphery nations, the dynamics of global news would undergo significant changes, as illustrated in Figure (9), which critically analyzes global connections by isolating the impact of these dominant countries.

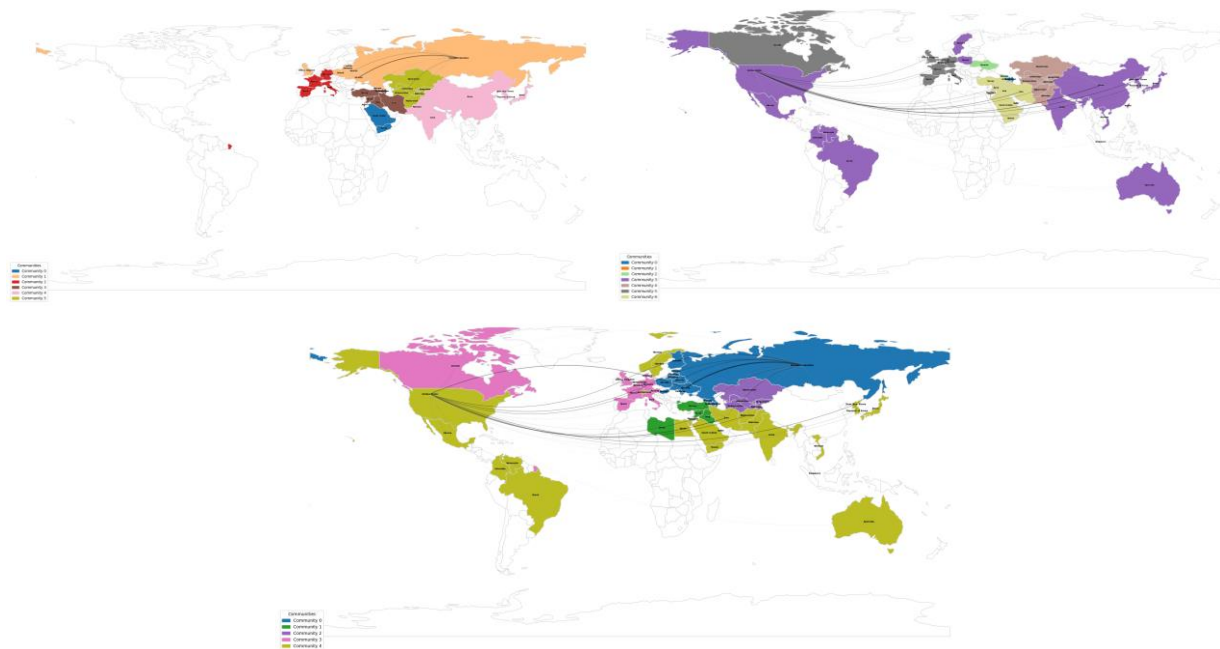


Figure 9. Core-Periphery Dynamics: Global Networks with Node Attack on the United States, Russian Federation, and China

By selectively focusing on node attacks targeting their connections, the figures reveal distinct patterns of regional and global dominance. When the United States was removed, its widespread and far-reaching network, which extensively connected across the Americas and reinforced its role as a global hegemon, disappeared. A node attack on the Russian Federation highlights a more regionally concentrated influence. Ongoing political conflicts, such as those with Ukraine and Georgia, have caused these nations to form separate communities. The removal of China, meanwhile, presents a shift in the balance of power, where regional dominance tilts toward the United States and the Russian Federation's strategic global outreach becomes more prominent.

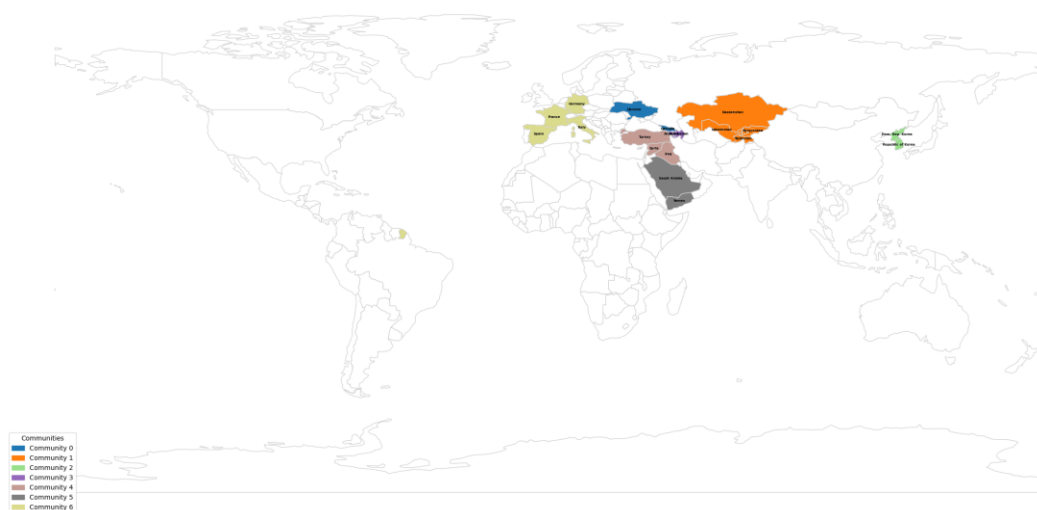


Figure 10. Global Network Fragmentation: Disconnecting Core Periphery Nations

When the three core nations, the United States, Russia, and China are removed from the global network, the structure becomes fragmented, breaking into smaller and disconnected clusters. Figure (10) reveals minimal interaction between peripheral regions, proving the reliance on core nations for global connectivity. The resulting map highlights regional clusters, such as connections within Central Asia, conflicts between the Koreas, European Union-related interactions, tensions in the Middle East, and conflicts in Turkey, Ukraine, and Georgia. This demonstrates that most international news flows are concentrated between the three core nations, with peripheral regions rarely exchanging information directly. According to the research done by Himelboim, the theoretical agreement is that the flow of foreign news is uneven, meaning some countries receive much more attention in global news than others¹⁴. Our argument is to prove the node attack of core nations reveals that peripheral countries lack significant interconnectivity and depend heavily on the core for news flow. This analysis proves that without the three core nations, international news coverage from and within peripheral regions largely disappears, underscoring the asymmetrical and centralized nature of global media networks.

These shifts emphasize the critical roles these core nations play in shaping global networks and their interactions. This targeted "node attack" approach underscores the asymmetric nature of core-periphery relationships in the global system, revealing the dependencies and hierarchies shaped by these dominant nations. It also highlights the resilience or fragmentation of the network when connections beyond these key nodes are removed. By conducting these node attacks, we gained deeper insights into the nature of affiliations between these core nations and their national interests on the global stage.

Conclusion

In the hierarchy of information flow, peripheral nations lack concentrated attention to attract readers, as foreign news presented by the Kun.uz platform strategically focuses on distant core countries, perpetuating global inequities in news coverage. However, it is not our aim to democratize the news flow but to explore the critical role of core nations in shaping global narratives and maintaining asymmetrical flows of information. Our research reaffirms the persistence of Wallerstein's World System Theory in modern media, and it also examines the network structure of core countries, overall findings highlighting the interconnectedness of nations and the enduring influence of core-periphery dynamics through the analysis of Uzbekistan's Kun.uz platform. Since the objective of our research aligns closely with the methodologies employed in Himelboim's study, the application of network analysis, Louvain community detection, and node attacks emerges as our central focus. These methods have revealed asymmetrical connectivity patterns, such as the peripheral countries remaining underrepresented and dependent on core nations for global news discourse. Node attacks, on the other hand, demonstrated the fragility of the international network, as the absence of key geopolitical powers significantly changed the direction of its structure and balance. These findings may emphasize the

¹⁴ Himelboim, Itai, Tsan-Kuo Chang and Stephen McCreery. "International Network of Foreign News Coverage: Old Global Hierarchies in a New Online World." *Journalism & Mass Communication Quarterly* 87 (2010): 297 - 314

importance of diversifying news coverage in developing countries to reduce the influence of the global powers. In the case of several regions such as Africa or Southeast Asia, global media landscapes are left with minor interest. Since global news continues to evolve, future research should explore how emerging methods and alternative platforms can reshape entrenched dynamics and promote more equitable and inclusive international news flows.

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